

Joint convexity of quantum relative entropy: literature context

References and proof-route comparison for `qRelativeEnt_joint_convexity`

This note places the formalized theorem and its proof route in the published literature, and records exactly which citations were verified while preparing this documentation (see §5). The result itself is classical; the route used in the Lean proof is assembled from standard published ingredients, as detailed below.

1. Joint convexity of the Umegaki relative entropy

The Umegaki relative entropy $D(\rho\|\sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)]$ was introduced by Umegaki [Umegaki62]. Its joint convexity,

$$D\left(\sum_i p_i \rho_i \parallel \sum_i p_i \sigma_i\right) \leq \sum_i p_i D(\rho_i \parallel \sigma_i),$$

is due to Lindblad [Lindblad74], as a consequence of Lieb’s concavity theorem [Lieb73] (joint concavity of $(A, B) \mapsto \text{Tr}[K^\dagger A^s K B^{1-s}]$, proved by Lieb to settle the Wigner–Yanase–Dyson conjecture). It is essentially equivalent to the strong subadditivity (SSA) of von Neumann entropy proved by Lieb and Ruskai [LiebRuskai73; LiebRuskaiPRL73], and it implies — via Lindblad [Lindblad75] and Uhlmann [Uhlmann77] — monotonicity of D under completely positive trace-preserving maps (the data-processing inequality). Later, structurally different proofs of joint convexity were given, e.g. via quasi-entropies and operator convexity (Petz [Petz86]), via operator perspective functions (Effros [Effros09]), and Tropp [Tropp12] showed the implication can be reversed (joint convexity $\Rightarrow$ Lieb’s theorem). A self-contained modern exposition of the classical route is Carlen’s lecture notes [Carlen10].

2. The sandwiched Rényi relative entropy

The sandwiched Rényi relative entropy

$$\tilde{D}_\alpha(\rho\|\sigma) = \frac{1}{\alpha - 1} \log \text{Tr}\left[\left(\sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}}\right)^\alpha\right]$$

was introduced independently by Müller-Lennert, Dupuis, Szehr, Fehr and Tomamichel [MLDSFT13] and by Wilde, Winter and Yang [WWY14]. Its key properties, as relevant here:

- **$\alpha \rightarrow 1$ limit.** $\tilde{D}_\alpha \rightarrow D$ (Umegaki) as $\alpha \rightarrow 1$; established in [MLDSFT13] (see also [WWY14]). This is the fact formalized as continuity of $\alpha \mapsto \tilde{D}_\alpha$ at $\alpha = 1$ (`sandwichedRelEntropy.continuousOn`).
- **Data processing.** Monotonicity under CPTP maps holds for all $\alpha \geq 1/2$. [MLDSFT13] proved it for a partial range of α and conjectured the general case; it was then proved for $\alpha > 1$ by Beigi [Beigi13] via complex interpolation, and by Frank and Lieb [FrankLieb13] for the full range $\alpha \geq 1/2$ via a variational argument. Leditzky, Rouzé and Datta [LRD17] revisit the Frank–Lieb variational proof to characterize equality; the Lean development of the sandwiched DPI (and of the joint convexity of the trace functional $\tilde{Q}_\alpha$) follows this Frank–Lieb/Leditzky–Rouzé–Datta presentation: variational formula $\tilde{Q}_\alpha = \sup_{H \geq 0} f_\alpha(H, \rho, \sigma)$, joint convexity of each $f_\alpha(H, \cdot, \cdot)$, twirling, and Stinespring dilation.
- **Operational meaning** for $\alpha > 1$ (strong converse exponents): Mosonyi and Ogawa [MosonyiOgawa15]; textbook treatments in Tomamichel [Tomamichel16] and Wilde [Wilde17].

At the matrix-analysis level, the convexity input behind the $\alpha > 1$ joint convexity of $\tilde{Q}_\alpha$ is the concavity of $\sigma \mapsto \text{Tr}[(\sigma^s H \sigma^s)^q]$ with $s = \frac{\alpha-1}{2\alpha}$, $q = \frac{\alpha}{\alpha-1}$ (note $sq = 1/2$, i.e. the functional is homogeneous of degree one in σ — the borderline case). Concavity statements of this type belong to the Epstein/Hiai family [Epstein73; Hiai13] and are the engine of the Frank–Lieb argument. In the Lean library this concavity (`HermitianMat.trace_conj_rpow_concave`) is derived from the Lieb and Ando trace theorems formalized in `QuantumInfo/ForMathlib/HayataGroup/TraceInequality/` [Lieb73; Ando79].

3. Comparison of proof routes

The classical route (Lindblad [Lindblad74], exposition in [Carlen10]): take Lieb’s jointly concave functional $(A, B) \mapsto \text{Tr}[K^\dagger A^s K B^{1-s}]$ and obtain joint convexity of D by a limit of difference quotients at the degenerate parameter value ($s \rightarrow 1$), where the functional reduces to a trace and its derivative produces the logarithms. This is the route taken, in one form or another, by most textbooks; the modern short proofs [Effros09; Tropp12] replace the limit by perspective-function arguments.

The route formalized here: take the *sandwiched* trace functional $\tilde{Q}_\alpha$, jointly convex for $\alpha > 1$ by the Frank–Lieb variational argument [FrankLieb13; LRD17], and pass to the limit $\alpha \rightarrow 1^+$, where $\tilde{D}_\alpha \rightarrow D$ [MLDSFT13]. The two elementary bounds $\log x \leq x - 1$ and $e^x \leq 1 + x + x^2$ (for $|x| \leq 1$) convert convexity of $\tilde{Q}_\alpha$ into an eventual convex-combination bound on $\tilde{D}_\alpha$ with a majorant that converges by continuity — so no differentiability at $\alpha = 1$ is needed.

Three honest observations about this comparison:

1. **The architecture is the same.** Both routes run: *jointly convex/concave one-parameter trace functional* $\rightarrow$ *limit at the degenerate parameter value where the relative entropy emerges as a derivative*. The formalized proof is the sandwiched analogue of Lindblad’s argument, with the Wigner–Yanase–Dyson-type functional replaced by $\tilde{Q}_\alpha$, and with a one-sided convergent majorant standing in for the derivative.
2. **The route is assembled from standard published pieces, but is not the textbook route to *this* theorem.** Joint convexity of the Umegaki entropy predates the sandwiched family by four decades, so the published literature invariably proves it by the classical route of §1. The $\alpha \rightarrow 1^+$ mechanism used here appears in the literature as the standard way to

transfer sandwiched-Rényi facts to the Umegaki entropy — e.g. the Umegaki DPI as the $\alpha \rightarrow 1$ limit of the sandwiched DPI [MLDSFT13; Wilde17], which is exactly how the library already proves `sandwichedRenyiEntropy_DPI_eq_one`. Applying the same limiting pattern to joint convexity is a routine application of the standard $\alpha \rightarrow 1^+$ transfer pattern; we do not claim a specific published reference whose stated result is “joint convexity of Umegaki via sandwiched $\alpha \rightarrow 1^+$ ”, and we are not aware of one.

3. **Both routes rest on the same analytic core.** Whether one goes through $\text{Tr}[K^\dagger A^s K B^{1-s}]$ or through $\tilde{Q}_\alpha$, the deep input is a Lieb-type trace concavity theorem [Lieb73; Ando79; Epstein73]. In the library, that core was already formalized (for the sandwiched DPI), which is why this route closes the open `sorry` with no new deep analysis: the contribution is the limit argument, the $[0, \infty]$ -valued case analysis, and the kernel/support bookkeeping.

4. Notes on citations inside the Lean source

Two pre-existing comments in `DPI.lean` (not introduced by this contribution) deserve a follow-up fix, flagged here for transparency:

- The module header (`DPI.lean:36`) says “Following Leditzky–Rouzé–Datta (arXiv:1306.5920)”. The name/eprint pairing is off: **arXiv:1306.5920 is Beigi** [Beigi13]; Leditzky–Rouzé–Datta is **arXiv:1604.02119** [LRD17]. Both eprint pages were checked directly on arXiv (2026-07-06) while preparing this note, so this is confirmed, not suspected. The *approach* actually followed (variational formula, twirling) is indeed Frank–Lieb as presented by Leditzky–Rouzé–Datta, so the names are right and the eprint number is wrong. We are happy to submit the one-token fix ($1306.5920 \rightarrow 1604.02119$) as a trivial follow-up if the maintainer would like it; it is deliberately *not* bundled into this PR to keep the PR single-concept.
- The docstring of `sandwichedTraceFunctional_jointly_convex` says “This is Proposition 3 of the paper, originally from Frank–Lieb”. This refers to the Leditzky–Rouzé–Datta presentation; the proposition number has not been re-verified against [LRD17] while preparing this note.

5. References

This section has two parts. The **verified references** below were checked against their arXiv abstract pages (and, where noted, the publisher’s listing) on 2026-07-06: title, full author list, eprint number, and the journal reference/DOI as shown there. These are citation-ready. After them is a short list of **standard background** — classic results every reader in the area knows — given by author/short-title/venue only, deliberately *without* volume/page/year, because those details have not been independently verified here and (per PhysLean’s AI policy) must be confirmed by the author before being cited.

None of these references is load-bearing for the correctness of the contribution. The correctness evidence is the Lean kernel (`#print axioms`, no `sorryAx`) and the build; this section is context only.

Verified references

- [Lindblad74] G. Lindblad, *Expectations and entropy inequalities for finite quantum systems*, Commun. Math. Phys. **39** (1974), 111–119. DOI [10.1007/BF01608390](https://doi.org/10.1007/BF01608390); also [Project Euclid](#). (Joint convexity of the relative entropy. Subscriber-access; not open — see [paper/original/README.md](#).)
- [Lindblad75] G. Lindblad, *Completely positive maps and entropy inequalities*, Commun. Math. Phys. **40** (1975), 147–151. DOI [10.1007/BF01609396](https://doi.org/10.1007/BF01609396). (Monotonicity under CP maps.)
- [MLDSFT13] M. Müller-Lennert, F. Dupuis, O. Szehr, S. Fehr, M. Tomamichel, *On quantum Rényi entropies: a new generalization and some properties*, J. Math. Phys. **54**, 122203 (2013). arXiv:1306.3142, doi:10.1063/1.4838856.
- [WWY14] M. M. Wilde, A. Winter, D. Yang, *Strong converse for the classical capacity of entanglement-breaking and Hadamard channels via a sandwiched Rényi relative entropy*, Commun. Math. Phys. **331** (2014), 593–622. arXiv:1306.1586, doi:10.1007/s00220-014-2122-x.
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- [Beigi13] S. Beigi, *Sandwiched Rényi divergence satisfies data processing inequality*, J. Math. Phys. **54**, 122202 (2013). arXiv:1306.5920, doi:10.1063/1.4838855. (Article number confirmed against the arXiv journal reference.)
- [LRD17] F. Leditzky, C. Rouzé, N. Datta, *Data processing for the sandwiched Rényi divergence: a condition for equality*, Lett. Math. Phys. **107** (2017), 61–80. arXiv:1604.02119, doi:10.1007/s11005-016-0896-9.
- [MosonyiOgawa15] M. Mosonyi, T. Ogawa, *Quantum hypothesis testing and the operational interpretation of the quantum Rényi relative entropies*, Commun. Math. Phys. **334** (2015), 1617–1648. arXiv:1309.3228, doi:10.1007/s00220-014-2248-x.
- [Tomamichel16] M. Tomamichel, *Quantum Information Processing with Finite Resources: Mathematical Foundations*, SpringerBriefs in Mathematical Physics vol. 5, Springer (2016). arXiv:1504.00233, doi:10.1007/978-3-319-21891-5. (Series volume 5 confirmed against the publisher listing.)

Standard background (not independently verified here)

Classic results in the area, listed by author / short title / venue only. Volume, page, and year are omitted deliberately: they have not been checked here and should be confirmed by the author before any of these is cited. None is relied on for correctness.

- [Umegaki62] H. Umegaki, *Conditional expectation in an operator algebra IV*, Kōdai Math. Sem. Rep.
- [Lieb73] E. H. Lieb, *Convex trace functions and the Wigner–Yanase–Dyson conjecture*, Adv. Math.
- [LiebRuskai73] E. H. Lieb, M. B. Ruskai, *Proof of the strong subadditivity of quantum-mechanical entropy*, J. Math. Phys.
- [LiebRuskaiPRL73] E. H. Lieb, M. B. Ruskai, *A fundamental property of quantum-mechanical entropy*, Phys. Rev. Lett.

- [Epstein73] H. Epstein, *Remarks on two theorems of E. Lieb*, Commun. Math. Phys.
- [Uhlmann77] A. Uhlmann, *Relative entropy and the Wigner–Yanase–Dyson–Lieb concavity...*, Commun. Math. Phys.
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- [Petz86] D. Petz, *Quasi-entropies for finite quantum systems*, Rep. Math. Phys.
- [Effros09] E. G. Effros, *A matrix convexity approach to some celebrated quantum inequalities*, Proc. Natl. Acad. Sci. USA.
- [Carlen10] E. A. Carlen, *Trace inequalities and quantum entropy: an introductory course*, Contemp. Math. (the Lean source’s “Theorem 2.10 of Carlen” refers to these notes).
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- [Hiai13] F. Hiai, *Concavity of certain matrix trace and norm functions*, Linear Algebra Appl.
- [Wilde17] M. M. Wilde, *Quantum Information Theory*, 2nd ed., Cambridge University Press.
- [Spohn78] H. Spohn, *Entropy production for quantum dynamical semigroups*, J. Math. Phys. (cited in the physics brief for the second-law connection).